YUE CAO

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PROFESSIONAL EXPERIENCE

- **Meta MSL Lab** *Mar. 2022 - Present*
  AI Research Scientist, Post Training

EDUCATION

- **Texas A&M University** *Sep. 2016 - Dec. 2021*
  Ph.D. in Electrical Engineering
- **University of Science and Technology of China** *Sep. 2012 - June 2016*
  B.S. in Applied Physics

RESEARCH INTERESTS & EXPERIENCE

Interests: My research focuses on agentic post-training, including multi-agents, vertical agentic applications, and AI for science.

Experience:

- **Avocado (Muse Spark)**: Release News
  - Tech lead for agentic financial ability, leading Muse Spark to **SOTA** on **Finance Agent v2** and **Tax Bench**.
  - Tech lead for Muse Spark's presentation quality, reaching reasoning Elo 1653 on **GDPval-AA v2** (latest version to be released), surpassing Opus 4.8 and Sonnet 5.0.
  - Tech lead for Muse Spark's model behaviour on steerability and proactiveness, making the model handle user interruption messages and subagent/device/cron messages better.
  - Core contributor to browsing agent and multi-agents, reflected by several benchmarks reported in the blog: **DeepSearchQA**, **WildSearch**, and multi-agent plots.
- **Llama3**: Core contributor to tool-use capabilities and safety.
- **AI for Science**: LLM for protein design and docking.

PUBLICATIONS

- A. Grattafiori and other authors including **Y. Cao**. *The Llama 3 Herd of Models*. **arXiv preprint** arXiv:2407.21783, 2024.
- Y. You, **Y. Cao**, T. Chen, Z. Wang, and Y. Shen. *Bayesian Modeling and Uncertainty Quantification for Learning to Optimize: What, Why, and How*. **International Conference on Learning Representations**, 2021.
- **Y. Cao**, P. Das, V. Chenthamarakshan, P. Chen, I. Melnyk, and Y. Shen. *Fold2Seq: A Joint Sequence(1D)-Fold(3D) Embedding-based Generative Model for Protein Design*. **International Conference on Machine Learning** 139, 1261-1271, 2021.
- **Y. Cao**, T. Chen, Z. Wang, and Y. Shen. *Learning to Optimize in Swarms*. **Advances in Neural Information Processing Systems** 32, 15018-15028, 2019.

- **Y. Cao** and Y. Shen. *TALE: Transformer-based protein function Annotation with joint sequence-Label Embedding*. **Bioinformatics** 37(18), 2825-2833, 2021.
- R. Taftaf and other authors including **Y. Cao**. *ICAM1 initiates CTC cluster formation and trans-endothelial migration in lung metastasis of breast cancer*. **Nature Communications** 12, 4867, 2021.
- **Y. Cao** and Y. Shen. *Bayesian Active Learning for Optimization and Uncertainty Quantification in Protein Docking*. **Journal of Chemical Theory and Computation** 16(8), 5334-5347, 2020.
- **Y. Cao** and Y. Shen. *Energy-based Graph Convolutional Networks for Scoring Protein Docking Models*. **Proteins: Structure, Function, and Bioinformatics** 88(8), 1091-1099, 2020.
- M. Karimi*, S. Zhu*, **Y. Cao*** and Y. Shen. *De Novo Protein Design for Novel Folds Using Guided Conditional Wasserstein Generative Adversarial Networks*. **Journal of Chemical Information and Modeling** 60(12), 5667-5681, 2020.
- **Y. Cao**, Y. Sun, M. Karimi, H. Chen, O. Moronfoye, and Y. Shen. *Predicting Pathogenicity of Missense Variants with Weakly Supervised Regression*. **Human Mutation** 40(9), 1579-1592, 2019.
- M. Kawaguchi, N. Dashzeveg, **Y. Cao**, Y. Jia, X. Liu, Y. Shen, and H. Liu. *Extracellular Domains I and II of cell-surface glycoprotein CD44 mediate its trans-homophilic dimerization and tumor cluster aggregation*. **Journal of Biological Chemistry** 295(9), 2640-2649, 2020.
- X. Liu and other authors including **Y. Cao**. *Homophilic CD44 Interactions Mediate Tumor Cell Aggregation and Polyclonal Metastasis in Patient-derived Breast Cancer Models*. **Cancer Discovery** 9(1), 96-113, 2019.
- M. S. Cline, G. Babbi, S. Bonache, **Y. Cao**, et al. *Assessment of blind predictions of the clinical significance of BRCA1 and BRCA2 variants*. **Human Mutation** 40(9), 1546-1556, 2019.
- M. Lensink and other authors including **Y. Cao**. *Blind prediction of Homo- and Hetero- Protein Complexes: The CASP13-CAPRI Experiment*. **Proteins: Structure, Function, and Bioinformatics** 87(12), 1200-1221, 2019.
- A. Voskanian and other authors including **Y. Cao**. *Assessing the Performance of in-silico Methods for Predicting the Pathogenicity of Variants in the Gene CHEK2 among Hispanic Females with Breast Cancer*. **Human Mutation** 40(9), 1612-1622, 2019.

AWARDS AND HONORS

- Received the NeurIPS Travel Award. Oct. 2019
- Received the NIH-funded CAGI Travel Fellowship. Nov. 2019
- Our team (Y. Cao and Y. Shen) ranked the 2nd among 26 groups for difficult targets in the 3rd joint CASP-CAPRI (Critical Assessment of protein Structure Prediction and Critical Assessment of PRedicted Interactions), a community-wide experiment on comparative evaluation of protein structure prediction and protein docking methods. Apr. 2019
- Our team (Y. Cao and Y. Shen) ranked the 3rd/51 for difficult targets in the 7th CAPRI (Critical Assessment of PRedicted Interactions), 2017-2019
- Received the First-class Award for Excellent Students in University of Science and Technology of China. Sep. 2015
- Bronze Medal in the 4th Asia-Pacific Informatics Olympiad. May 2010
- First-class Award in the National Olympiad in Physics in China. Nov. 2011
- First-class Award in the National Olympiad in Informatics in China. Nov. 2011
- First-class Award in the National Olympiad in Informatics in China. Dec. 2010

- First-class Award in the National Olympiad in Informatics in China. Dec. 2009

INVITED PRESENTATIONS

- **Y. Cao**, T. Chen, Z. Wang, Y. Shen. *Learning to Optimize in Swarms*. *(Poster)* **Advances in Neural Information Processing Systems (NeurIPS)**, Dec. 2019, Vancouver, Canada.
- **Y. Cao**, Y. Sun, M. Karimi, H. Chen, O. Moronfoye, and Y. Shen. *Predicting Pathogenicity of Missense Variants with Weakly Supervised Regression*. **Critical Assessment of Genome Interpretation (CAGI) Workshop**, Dec. 2019, San Francisco, USA.
- **Y. Cao** and Y. Shen. *Bayesian Active Learning for Optimization and Uncertainty Quantification in Protein Docking (Presented by Yang Shen)*. **Intelligent Systems for Molecular Biology (ISMB)**, July 2019, Basel, Switzerland.
- M. Karimi*, S. Zhu*, **Y. Cao*** and Y. Shen. *De Novo Protein Design of Novel Folds using Guided Conditional Generative Adversarial Networks (gcWGAN) (Poster)*. **Intelligent Systems for Molecular Biology (ISMB)**, July 2019, Basel, Switzerland.
- **Y. Cao** and Y. Shen. *Bayesian Active Learning for Optimization and Uncertainty Quantification in Protein Docking (Presented by Yang Shen)*. **7th CAPRI Evaluation Meeting**, April 2019, Hinxton, UK.
- **Y. Cao** and Y. Shen. *Bayesian Active Learning for Optimization and Uncertainty Quantification in Protein Docking (Poster)*. **Modeling of Protein Interaction (MPI)**, November 2018, Lawrence, KS, USA.
- **Y. Cao** and Y. Shen. *Bayesian Active Learning for Optimization and Uncertainty Quantification in Protein Docking (Poster)*. **Bioinformatics and Cancer Symposium**, Sep. 2018, College Station, TX, USA.

TECHNICAL SKILLS

Programming Languages: Python, C++, Bash Scripts
Deep Learning Frameworks: PyTorch, TensorFlow
Operating Systems: Linux, Mac OS, Windows
Other Computer Skills: Git, PyMOL, LaTeX, CHARMM